

duvall3's Fork of RAT-PAC

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A copy of the GPL is provided at `rat-pac/user/doc/README_duvall3/gpl-3.0.txt`

Simulation and Analysis Pipeline

NOTE: See *pipeline.pdf* in `user/doc/README_duvall3/` for a graphical representation.

Master Shell Script: `ratrun.sh`

For a `datarun` named `DATARUN`, `user/shell_scripts/ratrun.sh` will:

- Configure run by generating `run.mac`
- Create `DATARUN.conf` using `conflog.sh`
- Run `rat`, saving logs to `DATARUN.log` and ROOT data to `DATARUN.root`
- Run `process_rat_run.sh`:
 - Awk: `DATARUN.log -> DATARUN_energies.rt`
 - Run `duvallAnalyzeFull`:
 - * Run `rt_to_root`: `DATARUN_energies.rt -> DATARUN_energies_T.root`
 - * Run `particleTracksToScint`: `DATARUN.root -> DATARUN_T.root`
 - * Run `correctEnergies`: update `DATARUN_T.root`
 - * Run `duvallAnalyze`:

- Run `SEDAQ2`: create `DATARUN_results.root` and various PNG plots
- Run `angularRecon`: update `DATARUN_results.root` and create some more PNG plots
- Create datarun directory `DATARUN/` and move all related files to it
- If `GEOLOG` exists, copy it into `DATARUN/` as well

NOTE: This document assumes that a working installation of RAT-PAC and its accompanying build of ROOT are active on the machine in use. In other words, the version of ROOT used MUST be one compiled against RAT-PAC, as described in the RAT-PAC documentation [here](#).